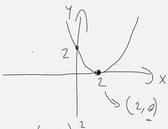
-1,



$f(x) = \frac{(x+k)^2}{2}$

$0 = \frac{(2+k)^2}{2} \quad | \cdot 2$

$(2+k)^2 = 0$

$2+k = 0$

$k = -2$

$f(x) = \frac{(x-2)^2}{2}$

$f(2)$

$f^{-1}(2)$

$\frac{(x-2)^2}{2} = 2$

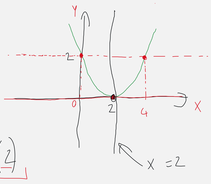
$\sqrt{4} = 2, -2$

$2^2 = 4 \quad (-2)^2 = 4$

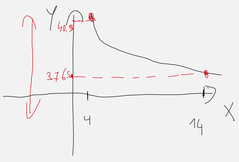
$(x-2)^2 = 4$

$x-2 = 2 \quad | \quad x-2 = -2$

$x = 4 \quad x = 0$



$f^{-1}(2)$



Range: 3.765, 40.5

[3.77, 40.5]

( )

$10 = \frac{640}{x^2} + 9.5$

$9.5 = \frac{640}{x^2} \quad | \cdot x^2$

$9.5x^2 = 640$

$x^2 = \frac{640}{9.5}$

$x^2 = 67.4$

$x_1 = 8.21$ $x_2 = -8.21$

$x \Rightarrow$ diameter

$h(x) \rightarrow$ height of can

$h^{-1}(10) = 8.21$

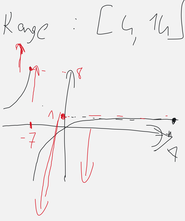
↳ For height of 10 the diameter 8.21

$h(10) = 8.21$

y-distance ↳ height

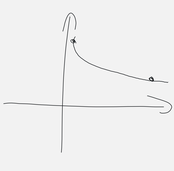
$4 \leq h^{-1}(x) \leq 14$

Range: $[4, 14]$



$y \leq 1$

$y \geq 8$



$0 = 2 - \frac{12}{x+5}$

$\frac{12}{x+5} = 2$

$12 = 2x + 10$

$x = 1$

$X = 0$